WORKED SOLUTION

STEP II 1998, Question 1

Pure • Diophantine Equations and Parity

STEP II 1998, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q1

The key idea

Expanding the cubes makes most of the terms cancel, leaving a cubic in n with a small constant term. Comparing parities on the two sides forces n to be even. Writing the cubic as $n^2(\text{something}) = 54$ then turns the question into one about divisors. For $(**)$, the even substitution $n = 2m$ takes us straight back to $(*)$.

Solution

The equation $(*)$

Using $(a+b)^3 - (a-b)^3 = 6a^2b + 2b^3$ with $a = n$, $b = 3$, we get

$$(n+3)^3 - (n-3)^3 = 18n^2 + 54,$$

so $(*)$ is equivalent to

$$n^3 = 18n^2 + 54, \qquad \text{that is,} \qquad n^2(n-18) = 54. \tag{1}$$

n is even. The right-hand side $18n^2 + 54 = 2(9n^2 + 27)$ is even for every integer n. If n were odd, then n^3 would be odd, contradicting $n^3 = 18n^2 + 54$. Hence n is even.

n^2 divides 54. In the second form of (1), $n - 18$ is an integer, so 54 is n^2 times an integer. Hence n^2 is a factor of 54.

Ask yourself How do these two facts together rule out every integer?

Since n is even, write $n = 2k$. Then $n^2 = 4k^2$, which is a multiple of 4 (and $n \neq 0$, since $n = 0$ gives $0 = 54$ in (1)). But $54 = 2 \times 27$ is not divisible by 4, so no multiple of 4 other than 0 can divide it. This contradiction shows that no integer n satisfies $(*)$.

(Another way to see it: the only squares dividing $54 = 2 \cdot 3^3$ are 1 and 9, so $n \in \{\pm 1, \pm 3\}$, all of which are odd.)

The equation $(**)$

Now take $b = 6$:

$$(n+6)^3 - (n-6)^3 = 36n^2 + 432,$$

so $(**)$ is equivalent to

$$n^3 = 36n^2 + 432.$$

The right-hand side is even for every integer n, so, exactly as before, n^3 is even. Since the cube of an odd number is odd, n is even.

Ask yourself Why can't we just copy the divisor argument from the first part?

We could write $n^2(n-36) = 432$, but $432 = 2^4 \cdot 3^3$ has plenty of even square factors (4, 16, 36, 144), so a divisibility argument alone doesn't finish. Instead we use the evenness. Put $n = 2m$ with m an integer. Then

$$8m^3 = 36 \cdot 4m^2 + 432 \quad \Longleftrightarrow \quad m^3 = 18m^2 + 54,$$

which is precisely equation (1) for the integer m. In other words, m satisfies $(*)$. Indeed, $(**)$ with $n = 2m$ is just $(*)$ with $n = m$, multiplied through by 8:

$$(2m-6)^3 + (2m)^3 - (2m+6)^3 = 8[(m-3)^3 + m^3 - (m+3)^3].$$

We showed that no integer satisfies $(*)$, so no such m exists. Hence no integer n satisfies $(**)$.

Checks

- Try $n = 1$ in the expansion: $(4)^3 - (-2)^3 = 64 + 8 = 72 = 18 + 54$. ✓
- The cubic $n^3 - 18n^2 - 54$ has exactly one real root, at about $n \approx 18.164$, which is not an integer. (A direct computer search over $|n| < 10^5$ finds no solution to either equation.)
- For $(**)$, the real solution is $n \approx 2 \times 18.164 = 36.33$, which matches the scaling $n = 2m$.

Common slips

- Forgetting that $n = 0$ needs a separate word: 0 is even, but 0 is not a factor of 54. Here (1) rules it out at once.
- Expanding the cubes in full and making an arithmetic error. The identity $(a+b)^3 - (a-b)^3 = 6a^2b + 2b^3$ halves the work.
- In $(**)$, trying to show $n^2 \mid 432$ leads nowhere. The point of "n is even" is the substitution $n = 2m$.

Beyond STEP

The final step, where a solution of $(**)$ produces a solution of an equation of the same shape with smaller numbers, is a simple form of Fermat's *method of descent*. Applied repeatedly, this method shows that $x^3 + y^3 = z^3$ has no solutions in positive integers, the cubic case of Fermat's Last Theorem first proved by Euler. Here the three terms form an arithmetic progression, which makes things much easier. In fact, $(n - 3k)^3 + n^3 = (n + 3k)^3$ says that $t = n/k$ is a root of $t^3 - 18t^2 - 54$, and by the rational root theorem this cubic has no rational roots, so there are no integer solutions for any $k \neq 0$. Davenport's *The Higher Arithmetic* is a lovely introduction to divisibility, parity arguments and Diophantine equations.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.